Module 03 | Sunday Quick Reference

Onsite: 9:00 AM-5:00 PM | Jupyter and Colab | Synthetic student data

01 Why Preprocessing Belongs in ML

Meaning: Real datasets may have missing numeric values, inconsistent categories and features measured in different units. Preprocessing converts them into usable inputs.

Remember: Training-only fitting matters: fill values, category mappings and scaling parameters must be learned on training rows, then reused on test and new rows.

Functions and terms: Imputation - Fill missing values using a learned rule; Encoding - Convert category labels into numbers; Scaling - Put numerical features on comparable scales

Common mistake: Never calculate missing-value replacements using the entire dataset before splitting; this can leak test information.

02 Missing Values with SimpleImputer

Meaning: SimpleImputer fills missing values with a learned statistic such as the median for numbers or the most frequent value for categories.

Remember: Median is often a helpful starting point when numeric values have unusual extremes. Missing values are not always random, so document the choice.

Functions and terms: strategy="median" - Use training-set numeric median; strategy="most_frequent" - Use most common training value; fit_transform() - Learn filling rule and apply it

Common mistake: Never assume that a missing cell means zero. Zero study hours and “unknown study hours” are different information.

03 Category Encoding

Meaning: One-hot encoding creates separate indicator columns for categorical values. With categories group, self and video, each category becomes a 0/1 column.

Remember: Encoding must learn the training categories and handle new, previously unseen categories safely. This is better than assigning arbitrary category numbers.

Functions and terms: OneHotEncoder - Convert categories into indicator columns; handle_unknown - Choose behavior for unseen categories; Sparse output - Compact representation of mostly zeros

Common mistake: Do not code group=1, self=2, video=3 unless those numbers have real ordered meaning.

04 Feature Scaling

Meaning: StandardScaler shifts each numeric feature around mean zero and scales it using standard deviation. It is helpful for models sensitive to feature magnitude.

Remember: Study hours may range 0–10, while attendance ranges 40–100. Scaling makes these comparable in algorithms such as logistic regression and K-means.

Functions and terms: Mean - Average of training feature values; Std. deviation - Spread used to divide centered values; transform() - Apply learned train-set scaling

Common mistake: Do not fit StandardScaler separately on the test data: this would make train and test use different transformations.

05 ColumnTransformer

Meaning: ColumnTransformer applies separate preprocessing steps to selected column groups, then joins the results into one numeric feature matrix.

Remember: This connects Pandas columns to a model safely. Numeric columns can receive imputation and scaling; text categories receive imputation and encoding.

Functions and terms: Numeric columns - Features needing imputation and scaling; Category columns - Features needing categorical encoding; ColumnTransformer - Combine transformers by named column lists

Common mistake: Do not manually encode the test data using a different set of categories or a different column order.

06 Pipeline - Link Preparation and Model

Meaning: A Pipeline chains preprocessing and a model. Calling fit() learns each training transformation and fits the estimator without using held-out test rows.

Remember: The same object handles predictions on new records, which reduces the risk of missing an encoding step when deploying the model.

Functions and terms: Pipeline steps - A named sequence ending with an estimator; fit() - Fit all transformers and final model; predict() - Transform new input and predict with same rules

Common mistake: The pipeline above requires prep and the train/test split from earlier. The full instructor notebook supplies and executes setup in order.

07 Feature Engineering

Meaning: Feature engineering creates new variables that express meaningful information. A well-designed feature may make a simple model more effective.

Remember: Create features only from information available at prediction time. New features should be tested on held-out data, not assumed better.

Functions and terms: Derived feature - A new column computed from existing inputs; Domain knowledge - Understanding how a feature relates to task; Ablation - Compare model with and without new feature

Common mistake: Creating the target from known final scores and using that computed feature to predict itself is leakage.

08 K-means Clustering

Meaning: K-means is unsupervised learning. It separates numeric records into k groups by repeatedly assigning each record to its nearest cluster center.

Remember: Clusters can help explore common patterns, but a cluster number is not a real-world diagnosis or a true label. Scaling matters because distance matters.

Functions and terms: k - Number of groups chosen before fitting; Centroid - Average location of a cluster; Label - Assigned cluster number, not ground truth

Common mistake: Clusters do not mean students belong to fixed personality types. K-means will assign groups even when the data lacks clear natural clusters.

09 Choosing k and Understanding Clusters

Meaning: The elbow method compares within-cluster inertia for several k values. Inertia usually decreases as k increases, so it is not itself proof of the best k.

Remember: Compare cluster centers and sizes, check whether the groups are meaningful, and avoid over-interpreting noisy results.

Functions and terms: Inertia - Sum of squared distances to assigned centers; Elbow - A point of diminishing improvement; Cluster profile - Typical feature values of a group

Common mistake: Do not claim a cluster represents successful students without comparing it against relevant outcomes and checking uncertainty.

10 Hyperparameter Tuning

Meaning: A hyperparameter is a setting chosen before training, such as tree depth. GridSearchCV tests a small list of settings using cross-validation.

Remember: Tuning should be performed inside the training portion only. Afterwards, the selected model is evaluated once on the reserved test set.

Functions and terms: Parameter - Value learned by fit, such as a coefficient; Hyperparameter - Choice made before fit, such as max_depth; GridSearchCV - Cross-validates candidate settings

Common mistake: Avoid tuning on X_test or repeatedly inspecting test performance after every grid. That quietly weakens the final evaluation.

11 Save, Load and Run New Predictions

Meaning: joblib saves fitted Python/scikit-learn objects to disk. Saving the entire Pipeline keeps the encoding/scaling and model together.

Remember: After loading, the model can run on new records without fitting again. New input must follow the original schema and feature meaning.

Functions and terms: Artifact - The saved fitted pipeline file; joblib.dump() - Write a fitted Python object to disk; joblib.load() - Read a trusted object back from disk

Common mistake: Only load joblib files from trusted sources; serialized Python objects can execute code. Keep library versions compatible.

12 Responsible ML and Model Cards

Meaning: Responsible use asks who the model affects, what data was used, whether results may be unfair, and which limitations are important to explain.

Remember: A model card records the model purpose, input columns, dataset, evaluation method, performance, limitations and conditions where it should not be used.

Functions and terms: Fairness - Check errors and impacts across groups; Privacy - Avoid unnecessary sensitive information; Model card - A short, reproducible model description

Common mistake: Do not feed names, private medical details or other unnecessary personal data into a student prediction model.